
purearray

Name

purearray, purearray-list, purearray-rename, purearray-replace, purearray-setattr — manages attributes that pertain to the array as a whole

purearray-monitor — monitors array I/O performance

purearray-disable, purearray-enable, purearray-phonehome, purearray-remoteassist — manages support-related functions

purearray-diagnose, purearray-test — performs array diagnostic functions

purearray-connect, purearray-disconnect — manages connections between arrays

Synopsis

`purearray connect --management-address MANAGEMENT-ADDRESS [--replication-address REPLICATION-ADDRESS] --type replication --connection-key`

`purearray diagnose { --email | --log | --phonehome }`

`purearray disable { console-lock | phonehome }`

`purearray disconnect ARRAY`

`purearray enable { console-lock | phonehome | security-token }`

`purearray list [--banner | --connect | --connection-key | --console-lock | --controller | --idle-timeout | --ntpserver | --phonehome | --proxy | --relayhost | --scsi-timeout | --security-token | --senderdomain | --space | --syslogserver | --throttle] [--path] [--historical { 1h | 3h | 24h | 7d | 30d | 90d | 1y }] [--cli | --csv | --nvp] [--notitle] [--page] [--raw]`

`purearray monitor [--csv] [ --historical { 1h | 3h | 24h | 7d | 30d | 90d | 1y } ] [--interval SECONDS] [--latency] [--mirrored] [--repeat REPEAT-COUNT] [--size] [ --notitle | --raw ]`

`purearray phonehome { --cancel | --send-all | --send-today | --send-yesterday | --status }`

`purearray remoteassist { --connect | --disconnect | --status }`

```
purearray rename NEW-NAME
```

```
purearray replace {security-token}
```

```
purearray setattr { --banner | --idle-timeout IDLE-TIMEOUT | --ntpserver NTP-SERVER | --proxy HTTPS-PROXY | --  
relayhost RELAY-HOST | --scsi-timeout | --senderdomain SENDER-DOMAIN | --syslogserver SYSLOG-SERVER }
```

```
purearray throttle --connect { --default-limit DEFAULT-LIMIT | --window WINDOW | --window-limit WINDOW-LIMIT }  
ARRAY
```

```
purearray test { ntp | pinghome | phonehome | security-token | syslogserver }
```

Options

-h | --help

Can be used with any command or subcommand to display a brief syntax description.

--banner

Displays or sets text that Purity users see in the GUI at login, and in the CLI just before the password prompt.

--cancel

Cancels any in-progress transmission of event logs to Pure Storage Support.

--connect

When used with **purearray list**, displays array connection details, including array name, management and replication addresses, connection status (true or false), throttle status (true or false), and connection type (for example, replication).

When used with **purearray remoteassist**, enables Pure Storage Support to initiate a remote assistance session.

For **purearray throttle**, must be used with any of the other `purearray throttle` options to set network bandwidth throttle limits.

--connection-key

When used with **purearray list**, displays a key that can be used to connect to this array from another array.

When used with **purearray connect**, prompts for the connection key of the remote array.

--controller

Displays the name, mode, FlashArray model, Purity//FA version, and status information for the array's controllers.

--default-limit *DEFAULT-LIMIT*

Sets the default maximum network bandwidth threshold for outbound traffic when transferring data to the specified target array. Once the threshold is exceeded, bandwidth throttling occurs.

--disconnect

Terminates an in-progress remoteassist session with Pure Storage Support.

--email

Sends subcommand output to Pure Storage Support.

--historical

When used with **purearray list --space**, displays the amount of usable physical storage on the array and the amount of storage occupied by data and metadata over the specified range of time. Valid time ranges include: 1 hour, 3 hours, 24 hours, 7 days, 30 days, 90 days, and one year.

When used with **purearray monitor**, displays historical performance data over the specified range of time. Valid time ranges include: 1 hour, 3 hours, 24 hours, 7 days, 30 days, 90 days, and one year.

--idle-timeout *IDLE-TIMEOUT*

Displays or sets an idle time limit in minutes. CLI and GUI sessions that are idle for more than *IDLE-TIMEOUT* minutes are automatically logged out. Values between 5 and 180 minutes are valid. Specifying a value of zero disables the automatic logout feature. The default timeout value is 30 minutes. Idle timeout changes apply to existing sessions.

--interval *SECONDS*

Sets the number of seconds between displays of real-time performance data. At each interval, the system displays a point-in-time snapshot of the performance data. If omitted, the interval defaults to every 5 seconds.

--latency

Displays real-time and historical I/O latency information across the entire array.

--log

Sends subcommand output to the array's log.

--management-address *MANAGEMENT-ADDRESS*

Specifies the management address. Enter the virtual (*vir0*) IP address or FQDN of the target array.

--mirrored

In a synchronous replication configuration, includes performance data for mirrored writes (i.e., writes to volumes in stretched pods). If one array of a stretched pod is offline, the writes processed on the other array are reported as local writes rather than mirrored ones. Once the pod is back online, in sync, and processing each write on both sides, the writes to its volumes are reported as mirrored writes again.

--ntpserver [*NTP-SERVER*]

Displays or sets the hostnames or IP addresses of the NTP servers currently being used by the array to maintain reference time.

When used with **purearray setattr**, specify up to four NTP servers. If specifying an IP address, for IPv4, specify the IP address in the form `ddd.ddd.ddd.ddd`, where `ddd` is a number ranging from 0 to 255 representing a group of 8 bits. For IPv6, specify the IP address in the form `xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`, where `xxxx` is a hexadecimal number representing a group of 16 bits. When specifying an IPv6 address, consecutive fields of zeros can be shortened by replacing the zeros with a double colon (`::`).

--path

Displays the connection paths between this array and each of the connected arrays. This option is only valid when run on the array from where the array connection was made. Must be used with the `--connect` option.

--phonehome

For **purearray diagnose**, sends subcommand output to Pure Storage Support via the phonehome channel.

For **purearray list**, displays the current state of the Purity//FA phonehome automatic hourly log transmission facility (enabled or disabled).

--proxy [*HTTPS-PROXY*]

Displays or sets the proxy host for the phonehome facility when https is the phonehome protocol (the phonehome facility itself determines which protocol to use). The format for the option value in the **purearray setattr** subcommand is: `HTTP(S)://HOSTNAME:PORT`, where `HOSTNAME` is the name of the proxy host, and `PORT` is the TCP/IP port number used by the proxy host.

--relayhost [*RELAY-HOST*]

Displays or sets the hostname or IP address of the email relay server currently being used as a forwarding point for email alerts generated by the array. If specifying an IP address, enter the IPv4 or IPv6 address.

For IPv4, specify the IP address in the form `ddd.ddd.ddd.ddd`, where `ddd` is a number ranging from 0 to 255 representing a group of 8 bits. If a port number is also specified, append it to the end of the address in the format `ddd.ddd.ddd.ddd:PORT`, where `PORT` represents the port number.

For IPv6, specify the IP address in the form `xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx`, where `xxxx` is a hexadecimal number representing a group of 16 bits. Consecutive fields of zeros can be shortened by replacing the zeros with a double colon (`::`). If a port number is also specified, enclose the entire address in square brackets (`[]`) and append the port number to the end of the address. For example,

`[xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx:xxxx]:PORT`, where `PORT` represents the port number.

--repeat *REPEAT-COUNT*

Sets the number of times to log I/O statistics or display real-time performance data. If omitted, the repeat count is 1.

--replication-address *REPLICATION-ADDRESS*

Specifies the replication address. Enter the IP address or FQDN of the replication bond (`replbond`) interface on the target array.

--scsi-timeout

Displays or sets the amount of time (in seconds) that can lapse during an I/O interruption before the target ports log out of the fabric. The default timeout value is 60 seconds.

The **--scsi-timeout** option is an advanced option. Changing the default timeout value may cause an initiator to mistakenly interpret the status of the FlashArray as failed or generate a host timeout.

Contact the Pure Storage support team before you change the **--scsi-timeout** value.

--security-token

Displays the status of the Rapid Data Locking feature (enabled or disabled) and the signature of the attached security token. For more information about the Rapid Data Locking (RDL) feature, refer to the FlashArray Enhanced Data Security Guide in the Pure1 Knowledge site at <https://support.purestorage.com>.

--send-all

Sends all log information stored in the array to Pure Storage Support via the phonehome channel.

--senderdomain [*SENDER-DOMAIN*]

Displays or sets the domain name. The domain name determines how Purity//FA logs are parsed and treated by Pure Storage Support and Escalations. The domain name is also appended to alert email messages. The default domain name is `please-configure.me`. If this is not a Pure Storage test array, set the sender domain to the actual customer domain name.

--send-today

Sends log information from the current day (in the array's time zone) to Pure Storage Support via the phonehome channel.

--send-yesterday

Sends log information from the previous day (in the array's time zone) to Pure Storage Support via the phonehome channel.

--size

Displays the average I/O sizes per read and write operation.

--space

Displays the amount of usable physical storage on the array and the amount of storage occupied by data and metadata.

Capacity

Total physical usable space on the array. Replacing a drive may result in a dip in usable capacity. This is intended behavior. RAID striping splits data across an array for redundancy purposes, spreading a write across multiple drives. A newly added drive cannot use its full capacity immediately but must stay in line with the available space on the other drives as writes are spread across them. As a result, usable capacity on the new drive may initially be reported as less than the amount expected because the array will not be able to write to the unallocatable space. Over time, usable capacity fluctuations will occur, but as data is written to the drive and spreads across the array, usable capacity will eventually return to expected levels.

Parity

Percentage of data that is fully protected. The percentage value will drop below 100% if the data isn't fully protected, such as when a module is pulled and the array is rebuilding the data to bring it back to full parity.

Thin Provisioning

Percentage of volume sectors that do not contain host-written data because the hosts have not written data to them or the sectors have been explicitly trimmed.

Data Reduction

Ratio of mapped sectors within a volume versus the amount of physical space the data occupies after data compression and deduplication. The data reduction ratio does not include thin provisioning savings.

For example, a data reduction ratio of 5:1 means that for every 5 MB the host writes to the array, 1 MB is stored on the array's flash modules.

Total Reduction

Ratio of provisioned sectors within a volume versus the amount of physical space the data occupies after reduction via data compression and deduplication *and* with thin provisioning savings. Total reduction is data reduction with thin provisioning savings.

For example, a total reduction ratio of 10:1 means that for every 10 MB of provisioned space, 1 MB is stored on the array's flash modules.

Volumes

Physical space occupied by volume data that is not shared between volumes, excluding array metadata and snapshots.

Snapshots

Physical space occupied by data unique to one or more snapshots.

Shared Space

Physical space occupied by deduplicated data, meaning that the space is shared with other volumes and snapshots as a result of data deduplication.

System

Physical space occupied by internal array metadata.

Total

Total physical space occupied by system, shared space, volume, and snapshot data.

--status

For **purearray phonehome --status**, displays the status of in-progress log transmission jobs. A status of "running" indicates a log transmission of the displayed action type (send-all, send-today, or send-yesterday) is in progress.

For **purearray remoteassist --status**, displays the status of a remote assist session as enabled or disabled.

--stop

Stops logging I/O statistics.

--syslogserver [*SYSLOG-SERVER*]

Displays or sets the URI of the remote syslog server. For example, <tcp://MyHost.com>.

Syslog servers deleted using this command will not receive an audit record on the remote syslog server. Instead, use the **purelog** command.

Note: The **--syslogserver** option is going to be replaced by the **purelog** command in a future release.

Therefore, **--syslogserver** is considered deprecated and should not be used. To set remote syslog servers, run the **purelog** command.

--throttle

Displays the network bandwidth throttle settings for all connected arrays. Must be used with the `--connect` option.

--type {replication}

The type(s) of connection desired. The only option supported in this version is 'replication'.

--window *WINDOW*

Range of time the `--window-limit` threshold is in effect.

--window-limit *WINDOW-LIMIT*

Sets the maximum network bandwidth threshold during the specified `--window` time range. Once the threshold is exceeded, bandwidth throttling occurs.

Options that control display format:

--cli

Displays output in the form of CLI commands that can be issued to reproduce the current configuration. The `--cli` output is not meaningful when combined with immutable attributes.

--csv

Lists information in comma-separated value (CSV) format. The `--csv` output can be used for scripting purposes and imported into spreadsheet programs.

--notitle

Lists information without column titles.

--nvp

Lists information in name-value pair (NVP) format, in the form `ITEMNAME=VALUE`. Argument names and information items are displayed flush left. The `--nvp` output is designed both for convenient viewing of what might otherwise be wide listings, and for parsing individual items for scripting purposes.

--page

Turns on interactive paging.

--raw

Displays the unformatted version of column titles and data. For example, in the `purearray monitor` output, the unformatted version of column title `us/op (read)` is `usec_per_read_op`. The `--raw` output is used to sort and filter list results.

Arguments

ARRAY

Name of the remote array.

NEW-NAME

Name by which the array is to be known after the command executes.

Object Names

Valid characters are letters (A-Z and a-z), digits (0-9), and the hyphen (-) character. The first and last characters of the name must be alphanumeric, and the name must contain at least one letter.

Most objects in Purity//FA that can be named, including host groups, hosts, volumes, protection groups, volume and protection group suffixes, SNMP managers, and subnets, can be 1-63 characters in length.

Array names can be 1-56 characters in length. The array name length is limited to 56 characters so that the names of the individual controllers, which are assigned by Purity//FA based on the array name, do not exceed the maximum allowed by DNS.

Names are case-insensitive on input. For example, `vol1`, `Vol1`, and `VOL1` all represent the same volume. Purity//FA displays names in the case in which they were specified when created or renamed.

Pods and volume groups provide a namespace with unique naming conventions.

All objects in a pod have a fully qualified name that include the pod name and object name. The fully qualified name of a volume in a pod is `POD::VOLUME`, with double colons (`::`) separating the pod name and volume name. The fully qualified name of a protection group in a pod is `POD::PGROUP`, with double colons (`::`) separating the pod name and protection group name. For example, the fully qualified name of a volume named `vol01` in a pod named `pod01` is `pod01::vol01`, and the fully qualified name of a protection group named `pgroup01` in a pod named `pod01` is `pod01::pgroup01`.

If a protection group in a pod is configured to asynchronously replicate data to a target array, the fully qualified name of the protection group on the target array is `POD:PGROUP`, with single colons (`:`) separating the pod name and protection group name. For example, if protection group `pod01::pgroup01` on source array `array01` asynchronously replicates data to target array `array02`, the fully qualified name of the protection group on target array `array02` is `pod01:pgroup01`.

All objects in a volume group have a fully qualified name that includes the volume group name and the object name, separated by a forward slash (`/`). For example, the fully qualified name of a volume named `vol01` in a volume group named `vgroup01` is `vgroup01/vol01`.

Description

The **purearray diagnose** subcommand executes a series of CLI subcommands that capture a snapshot of an array's I/O performance, configuration, and hardware status to be collected by Pure Storage Support. The output format must be declared with one of the following options:

--email

Mail the output to Pure Storage Support.

--log

Write the output to the array's log.

--phonehome

Transmit the output to Pure Storage Support via the secure channel.

The **purearray enable phonehome** and **purearray disable phonehome** commands enable and disable, respectively, automatic hourly transmission of array logs to Pure Storage Support.

The **purearray enable console-lock** and **purearray disable console-lock** commands enable and disable, respectively, root user login at the physical (serial or VGA) console.

The **purearray enable security-token** enables Rapid Data Locking if the required hardware is present. The command must be run on the primary controller.

The **purearray list** command with no option specified displays the array name, the array ID, and the Purity//FA version and revision numbers. Options can be specified to display controller information, connected arrays, the NTP (Network Time Protocol) server name, the state of and proxy host for the phonehome facility, the state of the console lock, the relay host via which electronic mail is sent, and the sender domain for electronic mail alert messages.

The **purearray list --connect** command displays a list of arrays that are connected to this array. In the list output, the management and replication addresses only appear for the arrays from where an array connection was made. If the array connection was made from its peer array, the Management Address and Replication Address columns display dashes (-). Include the **--path** option to display the connection paths for each of the connected arrays. The **--path** option is only valid when run on the array from where the array connection was made. A status of **connected** means that the path between the source and destination is healthy. A status of **connecting** means that the connection is unhealthy, possibly due to network issues. When a path goes into **connecting** status, the array generates an alert citing a degraded connection, which may result in replication issues. Check the network connectivity between the replication interfaces.

The **purearray list --controller** command displays the name, mode, FlashArray model, Purity//FA version, and status information for the array's controllers. The following are the controller modes:

primary

Running as the primary controller.

secondary

Running as the secondary controller.

not present

Not installed.

With FA-400 systems, also represents a controller that is installed but not running Purity//FA at the moment (including a controller that is powered off).

offline

Installed but not running Purity//FA at the moment. For example, represents a controller that is powered off or is undergoing a Purity//FA or firmware upgrade.

This status does not apply to FA-400 systems, where physical presence could not reliably be determined.

The Status column contains one of the following:

ready

Functioning normally.

not ready

In the process of starting Purity//FA.

updating

Undergoing a firmware upgrade. This status does not apply to FA-400 systems.

unknown

Status could not be determined.

For arrays to which a second controller has never been connected, only a single controller is displayed. Once an additional controller has been connected, its information is always shown.

The **purearray list --space** command displays current physical storage capacity, both total configured and occupied by data. Include the **--historical** option to display historical space data at the specified resolution.

The **purearray monitor** command displays real-time and historical I/O performance information for the whole array. The output includes the following data about bandwidth, IOPS, latency, and queue depth:

- **Time:** Current time.
- **B/s:** Bandwidth. Number of bytes read/written.
- **op/s:** IOPS. Number of read, write, or mirrored write requests processed per second.

- **us/op**: Service time. Average time, measured in microseconds, it takes the array to serve a read, write, or mirrored write I/O request. Service time does not include SAN time, queue time, or QoS rate limit time.
- **B/op**: IOPS. Average I/O size per read, write, mirrored write, and both read and write (all) operations. Must include the **--size** option to see the B/op columns.
- **Queue us/op**: Queue time. Average time, measured in microseconds, that a read, write, or mirrored write I/O request spends in the array waiting to be served.
- **Depth**: Queue depth. Average number of queued I/O requests for all volumes.

By default, the **purearray monitor** command displays real-time performance data. Include the **--repeat** option to specify the number of times to repeat the real-time update. If not specified, the repeat value defaults to 1. Include the **--interval** option to specify the number of seconds between each real-time update. At each interval, the system displays a point-in-time snapshot of the performance data. If not specified, the interval value defaults to every 5 seconds. The **--interval** and **--repeat** options can be combined.

In a synchronous replication configuration, include the **--mirrored** option to display performance data for mirrored writes (i.e., writes to volumes in stretched pods). If one array of a stretched pod is offline, the writes processed of the other array are reported as local writes rather than mirrored ones. Once the pod is back online, in sync, and processing each write on both sides, the writes to its volumes are reported as mirrored writes again.

The **purearray monitor --historical** command displays historical performance data over any of the following ranges of time: 1 hour, 3 hours, 24 hours, 7 days, 30 days, 90 days, or 1 year.

The **purearray monitor --latency** command displays real-time and historical I/O latency information across the entire array. The **purearray monitor --latency** output includes the following data:

- **Time**: Current time.
- **SAN us/op**: SAN time. Average time, measured in microseconds, required to transfer data between the initiator and the array. Slow data transfers will result in higher SAN times.
- **Queue us/op**: Queue time. Average time, measured in microseconds, that an I/O request spends in the array waiting to be served. The time is averaged across all I/Os of the selected types.
- **QoS Rate Limit us/op**: QoS rate limit time. Average time, measured in microseconds, that all reads, writes, or mirrored writes spend in queue as a result of bandwidth limits reached on one or more volumes.
- **us/op**: Service time. Average time, measured in microseconds, it takes the array to serve read, write, or mirrored write I/O requests across all volumes. Service time does not include SAN time, queue time, or QoS rate limit time.

Include the **--mirrored** option to display latency data for mirrored writes (i.e., writes to volumes in stretched pods).

The **purearray phonehome** subcommand initiates immediate transmission of event logs stored in an array to Pure Storage support. Options determine whether the current day, previous day, or all log information in the array are transmitted. Only one transmission may be in progress at any instant. A transmission in progress can be canceled by specifying the **--cancel** option in the subcommand. The phonehome facility automatically selects either the ssh or the https protocol for communicating with Pure Storage support. If a proxy host is required by https, use the **purearray setattr --proxy** command to specify its hostname and port. Specify the full URL string as described in the OPTIONS section above.

Purity//FA can transmit an array's log contents to Pure Storage Support (a) automatically once per hour, or (b) on demand. Pure Storage Support stores log contents for immediate or later analysis.

On-demand log transmission may be complete (**--send-all** option) or selective (**--send-today** or **--send-yesterday** options). Run the **--status** option to check the status of an in-progress log transmission. Run the **--cancel** option to cancel an in-progress log transmission.

Purity//FA can collect samples of hardware status, monitored performance, and array configuration on demand, and either (a) mail them to enabled alert recipients, (b) write them into the array's log, or (c) transmit them immediately to Pure Storage Support for analysis.

The **purearray remoteassist** subcommand enables or terminates a detached remote assistance session with Pure Storage Support. When the **purearray remoteassist --connect** command enables a session, a port ID is displayed. The administrator relays this ID to a Pure Storage Support representative, usually by phone, to enable the technician to connect to the array and perform diagnostic functions. The administrative session from which the **purearray remoteassist --connect** command is issued is unaffected, and can terminate the session at any time via the **purearray remoteassist --disconnect** subcommand.

One remote assistance session can be active at a time. Executing the **purearray remoteassist --connect** command while a session is active results in an error message. The **purearray remoteassist --status** command displays remote assistance status (capability disabled, capability enabled, or session active).

Use the **purearray rename** command to change an array's name. See the Object Names section below for information on Purity//FA object naming rules.

Use the **purearray setattr** command to change other attributes. To set a null value, use an empty string ("").

If Rapid Data Locking was enabled during **puresetup**, the **purearray replace** command substitutes new security tokens for those currently in use. Additional security tokens should be connected to both controllers prior to issuing the command. The command updates flash module keys using the passphrase on the new security tokens.

Purity//FA can test certain array functionality on demand. The **purearray test ntp** command verifies that the configured NTP servers can be used to synchronize time on both controllers with Coordinated Universal Time (UTC). The **purearray test pinghome** command tests an array's ability to connect with Pure Storage Support by establishing a secure shell or secure http connection. The **purearray test phonehome** command extends the **purearray test pinghome** command by additionally verifying that test messages can be exchanged. The **purearray test syslogserver** command sends a test message to configured syslog servers.

The **purearray test security-token** command tests the functionality of attached security tokens by regenerating and displaying an enhanced secret used to encrypt flash modules. The command verifies that attached security tokens are accessible and contain the passphrase used to generate the flash module keys currently in use. The command returns an error if:

- Rapid Data Locking is not enabled (configured during **puresetup**).
- The security token is not present or not functional.
- The attached security token does not contain the correct passphrase.

For more information about security tokens and the Rapid Data Locking (RDL) feature, refer to the FlashArray Enhanced Data Security Guide in the Pure1 Knowledge site at <https://support.purestorage.com>.

Sender Domain and Purity//FA Logs

The **purearray setattr --senderdomain** command determines how Purity//FA logs are parsed and treated by Pure Storage Support and Escalations. It is crucial that you set the sender domain to the correct domain name.

If the array is not a Pure Storage test array, set the sender domain to the actual customer domain name. For example, mycompany.com.

If the sender domain is set to the default please-configure.me domain name, the array will be treated as an internal Pure Storage test array, potentially blocking alerts and other important warnings.

Purity//FA also includes the sender domain name in the email addresses of outgoing alert messages. For example, purearray-ct0@mycompany.com.

Run **purearray list --senderdomain** to display the sender domain name.

Connecting Two Arrays

To transfer data from one array to another, the two arrays must be connected. For example, arrays must be connected to replicate data from one array to another.

Arrays are connected using a connection key, which is supplied from one array and entered into the other array.

When two arrays are connected, the array where data is being transferred from is called the source (local) array, and the array where data is being transferred to is called the target (remote) array. A source array can connect to multiple target arrays.

Connecting two arrays for data replication involves the following steps:

1. From the target array, obtain the connection key (**purearray list --connection-key**).

2. From the source array, connect the source array to the target array (`purearray connect --management-address --type --connection-key`).
3. From the source array, create the protection group (`purepgroup create`), configure the protection group replication schedule (`purepgroup schedule`), and enable the protection group replication schedule (`purepgroup enable`).

The **purearray connect --management-address --type --connection-key** command connects the local array to a remote array. The **--management-address** option represents the virtual (`vir0`) IP address or FQDN of the target array. The **--connection-key** option represents the connection key of the remote array. To obtain the connection key, run `purearray list --connection-key` from the remote array. The **-type** option represents the type of connection. Purity//FA supports connection type `replication`. Include the **--replication-address** option, which represents the IP address or FQDN of the replication bond (`replbond`) interface on the target array.

Once two arrays are connected, optionally configure network bandwidth throttling to set maximum threshold values for outbound traffic.

The **purearray disconnect** command disconnects the local array from the specified remote array. The command can only be run on the array that established the connection. Disconnecting two arrays suspends any in-progress data transfer processes. The process resumes when the arrays are reconnected.

If two arrays are connected for replication purposes, disconnecting the arrays does not delete data from the replication target array. However, after the two arrays have been disconnected, removing a target array from a protection group immediately deletes all of the data on the target array that was sent from its source array. As a result, replicating data again from the source array to the target array would cause another baseline transfer.

Network Bandwidth Throttling

Once two arrays are connected, optionally configure **purearray throttle --connect** to regulate when and how much data should be transferred between the arrays. The **purearray throttle --connect** command must be used with at least one of the following options: `--default-limit`, `--window-limit`, or `--window`.

Network bandwidth throttling on the source array sets a network bandwidth cap for the amount of outbound data, measured by number of bytes per second, that can be transferred from the local array to a target array before network bandwidth throttling occurs.

Two different network bandwidth limits can be set:

- The **--default-limit** option sets a default maximum network bandwidth threshold for outbound traffic.

and/or

- The **--window-limit** option sets a maximum network bandwidth threshold for outbound traffic during the specified **--window** time range.

If both thresholds are set, the “window” limit overrides the “default” limit.

The maximum threshold must be between one megabyte per second and four gigabytes per second. The maximum threshold value represents an average data rate, so actual data transfer rates can fluctuate slightly above the configured limit.

The `--window-limit` setting overrides the `--default-limit` setting.

The `--default-limit` and `--window-limit` values are specified as integers, followed by one of the suffix letters `M` or `G`, denoting 512-byte sectors MiB and GiB, respectively.

The `--window` start and end times must be set on the hour, and the time can be specified in 12-hour or 24-hour format.

For example, run the following command to set a default network bandwidth cap of 4 gigabytes per second of data to be transferred from the local array to array `ARRAY1`, except between the hours of 11:00am and 3:00pm, when the network bandwidth limit drops to 2 gigabytes per second.

```
$ purearray throttle --connect --default-limit 4gb  
--window 11am-3pm --window-limit 2gb ARRAY1
```

To clear the `--default-limit`, `--window-limit`, or `--window` settings, use an empty string (“”).

To completely stop the data transfer process, set the `default-limit` or `window-limit` option to “0”. During this time, all in-progress and scheduled data transfer processes are aborted.

For example, run the following command to stop all data transfers from the local array to `ARRAY1` between 9:00am and 5:00pm.

```
$ purearray throttle --connect --window 9am-5pm --window-limit 0 ARRAY1
```

Bandwidth limit changes take effect immediately.

Note on Array Time

The installation technician sets the proper time zone for an array when it is installed. During operation, arrays maintain time synchronization by interacting with a Network Time Protocol (NTP) server (by default, time.purestorage.com). The **`purearray setattr --ntpserver NTP-SERVER`** command can be executed to specify an alternate NTP server by IP address or hostname.

Examples

Example 1

```
purearray enable phonehome
```

Enables automatic hourly transmission of log contents to the Pure Storage Support web site.

Example 2

```
purearray diagnose --phonehome
```

Takes a snapshot of performance, configuration, and hardware status and transmits it to Pure Storage Support via the phonehome channel.

Example 3

```
purearray list --phonehome
```

Displays the current state of the phonehome automatic hourly log transmission facility (enabled or disabled).

Example 4

```
purearray list --space --historical 3h --csv
```

Displays a CSV output of the amount of usable physical storage on the array and the amount of storage occupied by data and metadata over the past 3 hours.

Example 5

```
purearray list --throttle --connect
```

Displays network bandwidth limits, if set, for all connected arrays.

Example 6

```
purearray monitor --repeat 20 --interval 10
```

Displays real-time performance data for the whole array. Twenty (20) point-in-time updates are displayed, with each update taken every ten (10) seconds.

Example 7

```
purearray monitor --historical 24h --csv
```

Displays a CSV output of historical performance data for the local array over the past 24 hours.

Example 8

```
purearray phonehome --send-today
```

Initiates immediate transmission of the current day's event logs to Pure Storage Support via the phonehome network channel.

Example 9

```
purearray remoteassist --connect
```

Enables a Pure Storage Support representative to initiate a remoteassist session with the array.

Example 10

```
purearray rename NEWARRAYNAME
```

Changes the name of an array to [NEWARRAYNAME](#).

Example 11

```
purearray setattr --ntpserver MyNTPServer1.com,MyNTPServer2.com
```

Assigns the NTP servers [MyNTPServer1.com](#) and [MyNTPServer2.com](#) as the array's sources for reference time. Supersedes any previous NTP server assignments.

Example 12

```
purearray setattr --relayhost ""
```

Sets the relayhost attribute value to null, causing Purity//FA to send alert email messages directly to recipient addresses rather than routing them via a relay (mail forwarding) server.

Example 13

```
purearray setattr --relayhost fake.relay.purestorage.com:26
purealert test example@purestorage.com
```

Sets up a non-standard SMTP relay port number and sends a test alert.

Example 14

```
purearray setattr --senderdomain mycompany.com
```

Sets the sender domain name to `mycompany.com`.

Example 15

```
purearray list --connection-key
```

Displays the connection key of the array.

Example 16

```
purearray connect --management-address 10.78.0.188 --type replication
--connection-key 6207d123-d123-0b5c-5fa1-95fab5c7123
```

Connects the local array to remote array `0.78.0.188` using connection key `6207d123-d123-0b5c-5fa1-95fab5c7123` (taken from array `10.78.0.188`).

Example 17

```
purearray disconnect 10.78.0.188
```

Disconnects array `10.72.0.188` from the local (current) array.

Example 18

```
purearray throttle --connect --default-limit 2mb
--window 11pm-5am --window-limit 4mb ARRAY1
```

Sets the default maximum network bandwidth limit for outbound traffic to 2 MB/s for array `ARRAY1`, and sets the maximum network bandwidth limit for outbound traffic to 4 MB/s between 11:00pm and 5:00am.

See Also

[purealert\(1\)](#), [puredns\(1\)](#)

[purehgroup\(1\)](#), [purehost\(1\)](#), [purepgroup\(1\)](#), [purepod\(1\)](#), [purevol\(1\)](#)

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